

Illinois Environmental Protection Agency
Bureau of Water Document Control Number 128

Standard Operating Procedure for Continuous Monitoring of Water Quality

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1. Scope and Application

- 1.1. This Standard Operating Procedure (SOP) addresses the use of specialized equipment to continuously monitor several parameters in wadable streams. Future revisions of this SOP may address monitoring in larger streams and in lakes.
- 1.2. This SOP applies within the broader context of Appendix 1, Section B of the *Quality Assurance Project Plan: Integrated Water Monitoring Program Document* (Illinois EPA 1994), which describes sample-collection methods, and calibration and operation procedures for water-monitoring instruments.

2. Summary of Method

Qualified persons place a specialized monitoring device (hereafter, called a *sonde*) at specified sites in wadable streams for extended periods of time (typically seven days). The sonde measures and records physicochemical conditions in the water column, such as pH, specific conductance, temperature, and dissolved oxygen, at intervals defined by the user (e.g., once every 30 minutes). Each sonde has multiple *sensors*, each designed to measure a particular parameter. Measuring at such relatively short intervals over an extended period is called *continuous monitoring*. For clarity and brevity, this SOP will refer to each extended period of continuous monitoring as a *monitoring period*.

3. Definitions

- 3.1 A *sonde* is a group of sensors configured together with a recording unit or electronic data logger to record the output of multiple sensors.
- 3.2 A *sensor* is the part of the sonde that measures a particular parameter.
- 3.3 *Calibration drift* refers to changes in the accuracy of sensor/instrument calibration during the monitoring period. Section 11 addresses how to quantify calibration drift.
- 3.4 *Fouling* is surface contamination of sensors that results in inaccurate measurement attributable to chemical, physical, or biological factors such as drastic changes in flow, chemical contamination, excessive physical abrasion or inundation by sediment, or excessive growth of algae or bacteria.

4. Interferences and Corrective Action

- 4.1 Inaccuracy of sensor measurements obtained during the monitoring period is attributable to two sources: fouling and calibration drift. For each parameter in each monitoring period, users quantify this "total inaccuracy" by comparing measurements obtained with a second sonde or other appropriate instrument(s) to corresponding measurements

obtained by the continuously monitoring sonde. For each parameter in each monitoring period, users also quantify calibration drift. Consequently, inaccuracy attributable to fouling can be determined as the difference between total inaccuracy and inaccuracy attributable to calibration drift (see Section 11 for further details).

- 4.2 Theft and vandalism of monitoring instruments and equipment are potential problems. To help minimize these problems, users place sondes in inconspicuous locations in streams away from bridges or other access points, to the extent possible. To inhibit theft, instruments may be locked in a protective enclosure or cabled to fixed structures, (e.g., tree trunks).

5. Safety

Monitoring surface waters involves risk of harm to participants. Follow health and safety guidelines in DCN 151: *Surface Water Section Field Safety Manual* (Illinois EPA 2009, draft available on the Bureau of Water intranet or from the Bureau of Water Quality Assurance Officer).

6. Qualifications and Training of Personnel

Continuous monitoring is conducted by persons who have received formal training from qualified and experienced users of the procedures. Qualifications include the demonstrated ability to independently conduct the monitoring activities described in this document.

7. Equipment and Supplies

- 7.1. One or more sondes, each having a data recorder and one or more sensors for measuring water parameters such as pH, specific conductance, temperature, or dissolved oxygen.
- 7.2. A hand-held communications device or laptop computer for operating the sonde and downloading data from the sonde.
- 7.3. A protective housing with mounting frame, a cable, and a lock to contain and secure each sonde (Figure 1).
- 7.4. An additional sonde or other appropriate instrument(s) for obtaining independent measurements of water conditions at the end of the monitoring period.
- 7.5. Continuous Monitoring Log Sheets (Appendix A).
- 7.6. Materials (e.g., sample bottles) for collecting water samples.

- 7.7 Equipment for discharge monitoring; see Appendix 1, Section E of the *Quality Assurance Project Plan: Integrated Water Monitoring Program Document* (Illinois EPA 1994).
- 7.8 Digital camera for documenting general area and channel unit of sonde placement.
- 7.9 Geographical positioning system (GPS) unit for recording the exact location of sonde placement.



Figure 1. Sonde being placed into a protective housing mounted to a frame that suspends the sonde at a chosen depth.

8. Maintenance of Instruments and Equipment

Before each use (i.e., a single monitoring period), thoroughly clean all surfaces of each sonde. Replace the membrane of polarographic (i.e., Clark cell) dissolved oxygen sensors, and clean the surfaces of optical dissolved oxygen and other sensors, as needed, according to instructions in the instrument manufacturer's operating manuals and in Appendix 1, Section B, of *Quality Assurance Project Plan: Integrated Water Monitoring Program Document* (Illinois EPA 1994). Clean and lubricate all seals and connections of instruments and equipment. Remove sediment and other contaminants from the protective sonde housing by rinsing with clean water between uses.

9. Sample Preservation and Holding Time

Sample preservation and maximum allowable holding times do not apply to measurements from continuous monitoring instruments. For preservation and maximum

allowable holding times for water samples associated with the continuous monitoring, refer to Appendix 1, Section B, of *Quality Assurance Project Plan: Integrated Water Monitoring Program Document* (Illinois EPA 1994).

10. Monitoring Procedures

10.1. Sonde installation

Where and when to install the sonde in the stream and for how long of a period depend on the objectives of the continuous monitoring. For example, empirical evidence from various Illinois wadable streams indicates that dissolved oxygen (and possibly other parameters) can vary among channel units (i.e., riffles, runs, pools) in a stream; therefore, dissolved oxygen results can be affected by the placement location of the sonde in the monitored stream reach. Temperature and other parameters may vary with depth or location (e.g., proximity to a spring or to a discharge from a point source) within a stream reach. Currently, whenever possible under the individual circumstances, Illinois EPA installs the sonde in a run (channel unit) at a depth two-thirds of the distance from the water surface to the stream bottom. This placement is presumed to represent the average prevailing conditions a stream reach. The following procedures reflect Illinois EPA's current objectives for continuous monitoring, which are subject to change.

- 10.1.1. Monitor for two seven-day periods, one period during June 1 through July 31 and the other period during August 1 through October 15. A "day" means any 24-hour period, not necessarily a calendar day.
- 10.1.2. Before deployment calibrate each sensor on the sonde according to procedures in the instrument manufacturer's manuals and Appendix 1, Section B of *Quality Assurance Project Plan: Integrated Water Monitoring Program Document* (Illinois EPA 1994). Record all calibration information, including date and time of calibration, on the Continuous Monitoring Log Sheet for that instrument.
- 10.1.3. Place the sonde, preferably in a run (channel unit) in water that is at least one foot deep. For stream depths less than 1.5 ft suspend the sonde 6 inches above the stream bottom with the sensors facing downstream to minimize chances of damage from debris carried by stream current. As an exception, sondes with polarographic (i.e., Clark cell) dissolved oxygen sensors are faced upstream. For stream depths 1.5ft and greater, suspend the sonde two-thirds of the distance from the water surface to the stream bottom.
- 10.1.4. Set the frame in the stream bottom, place the sonde into the protective housing, and suspend the housing at the appropriate depth on the frame. Attach a cable to the assembly and fasten the cable to a fixed object such as a tree or a stake. Secure the cable and the assembly with locks.

- 10.1.5. If there is no U.S. Geological Survey gage at the continuous monitoring location, then measure stream discharge following the procedures outlined in Appendix 1, Section E of the *Quality Assurance Project Plan: Integrated Water Monitoring Program Document* (Illinois EPA 1994). Care should be taken to ensure that activities associated with the discharge measurement do not impact water quality sampling or the readings collected for the continuous monitoring.
- 10.1.6. Collect water samples before, during, or after the monitoring period, as appropriate. Care should be taken to ensure that activities associated with this sampling do not impact the readings collected for the continuous monitoring.
- 10.1.7. Complete the Continuous Monitoring Log Sheet, recording at least the following information:
 - monitoring location/station (e.g., Sugar Creek at U.S. Route 23 crossing; station XY-03)
 - location of sonde at site (i.e., channel unit, placement depth)
 - profile drawing of cross section of wetted channel with cross-section dimensions (e.g., wetted width, avg. depth) showing placement site of sonde at start of monitoring period
 - start (of monitoring period) date and time
 - end (of monitoring period) date and time
 - time interval of sonde readings (e.g., one reading per every 0.5 hours)
 - channel cross-section discharge at start and end of monitoring period
 - identification number of sonde (IEPA tag number)
 - miscellaneous comments

10.2. Sonde retrieval

- 10.2.1 At the end of the monitoring period, perform simultaneous measurements for each parameter, retrieve the sonde from the stream, download the monitoring data to a laptop computer or data logger, and complete post-deployment calibration check for drift and sensor fouling (Section 11).
- 10.2.2 If there is no U.S. Geological Survey gage at the continuous monitoring location, then measure stream discharge.

11. Quality Control and Corrective Action

- 11.1. Record on the appropriate Continuous Monitoring Log Sheet any deviations from the procedures in Section 10.
- 11.2. At the end of the monitoring period, for each parameter, quantify the total inaccuracy of the measurements obtained during the monitoring period. To do so, use a second sonde or other appropriate instrument(s) to independently measure each relevant

parameter in the water as close as possible to the sensors of the continuously monitoring sonde before it is retrieved from the water.

Obtain these independent measurements within 5 minutes of the final time interval in the monitoring period. On the Continuous Monitoring Log Sheet, record the independent measurement for each parameter and the time that it was obtained. When reviewing the results of continuous monitoring, compare the end-time independent measurement of each parameter to a corresponding result obtained by the continuously monitoring sonde. For each parameter, the total inaccuracy of the continuously monitoring sonde is quantified as the difference between the end-time independent measurement and the corresponding measurement from the sonde.

- For each parameter, if the *relative percent difference* (see formula below) between the two measurements is greater than 20%, then add one of two result qualifiers to every measurement of that parameter obtained during the entire monitoring period.
- Add an "F" (i.e., failed) qualifier to results judged to be inaccurate.
- A "Z" qualifier is used instead of an "F" qualifier, as follows: If the persons(s) who conducted the continuous monitoring can identify an obvious time during the monitoring period at which measurements became inaccurate (because of sensor fouling, low battery voltage, or other causes), then measurements prior to this time can be qualified with a "Z" instead of an "F". The "Z" qualifier means: "Although the post-use accuracy check failed the 20% *relative-percent-difference* criterion, additional evidence and insight by the person(s) who performed the continuous monitoring indicate that the result was likely obtained before the inaccuracy occurred in later measurements. Therefore, the result is likely sufficiently accurate." If a "Z" qualifier cannot be justified for a subset of the results, then assign an "F" qualifier to all results for those parameters for which the *relative percent difference* is greater than 20%.

Relative percent difference is the absolute value of the difference between two numbers (X_1 and X_2) divided by their mean and expressed as a percentage:

$$\text{Relative percent difference} = \{ (|X_1 - X_2|) / [(X_1 + X_2)/2] \} \times 100$$

- 11.3. At the end of each monitoring period, for each of pH, specific conductance and dissolved oxygen, determine the calibration drift. To do so, remove the sonde from the stream and rinse the sensors well with deionized water. Perform each appropriate post-retrieval calibration check on the sonde, but do not recalibrate the sensors (i.e., make no adjustments to the instruments). Rather, simply record (on the Continuous Monitoring Log Sheet) the reading when measuring the concentration of the calibration "standard" for each parameter. For each parameter during the monitoring

period, calibration drift is quantified as the difference between the "*Post-retrieval reading*" (Continuous Monitoring Log Sheet) and the "*Post-calibrated reading*" (Continuous Monitoring Log Sheet) of the pre-installation calibration of the sonde.

For pH, specific conductance, and dissolved oxygen measurements obtained during the monitoring period and not already qualified as "F" (failed) based on total inaccuracy (see section 11.2), determine acceptability by comparing the calibration drift of each parameter to critical thresholds as follows. Results of continuous monitoring are acceptable if calibration drift is no greater than the lesser of the following:

- two times the manufacturer's specified precision of the sensor/instrument, or
- the amounts in Table 1.

Table 1. Criteria for Data Acceptability

Measured Parameter	Criterion for Acceptability
pH	≤ 0.4 standard unit
Specific Conductance	$\leq 10\%$ *
Dissolved Oxygen	≤ 0.4 mg/L

* Relative percent difference (see section 11.2) between pre-installation and post-retrieval calibration readings.

Results of continuous monitoring are unacceptable if calibration drift does not meet the above criteria. For each parameter for which results are unacceptable, add an "F" (failed) qualifier to every result obtained during the monitoring period.

11.4 Periodically (once per year recommended), check the accuracy of each temperature sensor by comparing readings with those of a certified thermometer traceable to the National Institute of Standards and Technology.

12. Data Acquisition, Calculations, and Data Reduction

After retrieval, return sondes to the appropriate Monitoring Unit office, and use appropriate software to download the results of the continuous monitoring to an appropriate computer file (e.g., spreadsheet). Document and store these computer files to facilitate later upload to the appropriate Illinois EPA databases, taking all necessary steps to maintain data integrity.

13. Data and Records Management

The originals of all Continuous Monitoring Log Sheets are retained by the appropriate Monitoring Unit office for at least five years after each continuous monitoring period. Scanned copies of Continuous Monitoring Log Sheets are sent to Illinois EPA Bureau of Water headquarters for electronic attachment to continuous monitoring data files.

14. References

- Florida Department of Environmental Protection. 2004. *Continuous Monitoring with Installed Meters*. FT 1900. Tallahassee, FL.
- Hydrolab. 1991. *Operating Manual for Multiparameter Water Quality Data System*. Hydrolab Corporation. Austin, TX.
- Groschen, G.E. and R.B. King. 2005. *Evaluation of Measurements Collected with Multi-Parameter Continuous Water Monitors in Selected Illinois Streams, 2001-03*. Scientific Report 2005-5060. United States Geological Survey. Reston, VA.
- Illinois Environmental Protection Agency. 1994. *Quality Assurance Project Plan: Integrated Water Monitoring Program Document*. Bureau of Water, Springfield, IL.
- Johnston, M.W. and J.S. Williams. 2006. *Field Comparison of Optical Clark Cell Dissolved Oxygen Sensors in the Tualatin River, Oregon, 2005*. Open File Report 2006-1047. United States Geological Survey. Reston, VA.
- United States Geological Survey. 1998. *National Field Manual for the Collection of Water Quality Data*. Book 9, Handbooks for Water-Resource Investigations. United States Geological Survey. Reston, VA.
- Wagner, R.J., R.W. Boulger Jr., C. J. Oblinger, and B.A. Smith. 2006. *Guidelines and Standard Operating Procedures for Continuous Water-Quality Monitors: Station Operation, Record Computation, and Data Reporting*. Techniques and Methods 1-D3. United States Geological Survey. Reston, VA.

APPENDIX A

IEPA Continuous Monitoring Log Sheet

Station Code: _____		Stream name: _____	
Station Description: _____			
Lat _____		Long _____	

Pre-deployment Calibration of Sonde							
Date: _____	Time: _____ (24H CST)		Sonde Tag #: A				
BP (mm Hg): _____	Cup Temp (°C): _____	Battery: _____ IBV		Technician: _____			
	DO mg/l	Sp Cond zero	Sp Cond (µS/cm)	Sp Cond (µS/cm)	pH 7 Units	pH 10 Units	
Standard value: Pre-calibrated Post-calibrated Exp. Date/lot #							

Installation of Sonde							
Date: _____	Staff: _____						
Sonde Start Time: _____ (24H CST)	Sonde Time Interval (min): _____						
Placement Site of Sonde: _____							
GPS: N _____	W _____		± _____				
Stream Depth at Placement Site of Sonde _____ (ft)		Placement Depth of Sonde: _____ (ft) below surface					
~2/3 depth below surface _____		or _____		6 inch from bottom _____			
Channel Unit of Sonde Placement (check one): _____ riffle _____ run _____ pool _____							
Air Temp(°F): _____	% Cloud Cover: _____		Precip: _____				
Discharge: _____		Width (ft)	Avg. Depth (ft)	Velocity (fps)	Discharge (cfs)		

Retrieval of Sonde							
Date: _____	Sonde End Time: _____ (24H CST)		Staff: _____				
Air Temp(°F): _____	% Cloud Cover: _____		Precip: _____				
Discharge: _____		Width (ft)	Avg. Depth (ft)	Velocity (fps)	Discharge (cfs)		

Independent Measure of Physiochemical Conditions on End Date of Monitoring Period							
Time (24H CST) : _____	Water Temp (°C) _____	pH (units) _____		SpC(µS/cm) _____			
DO (mg/L) _____	Turbidity(NTU) _____		Sonde Tag #: A				

Post-retrieval Calibration Check of 7-Day Sonde							
Date: _____	Time: _____ (24H CST)		BP (mm Hg): _____		Cup Temp (°C): _____		
Battery: _____ IBV	DO mg/l	Sp Cond zero	Sp Cond (µS/cm)	Sp Cond (µS/cm)	pH 7 Units	pH 10 Units	
Standard value: Post-retrieval Exp. Date/lot #							

Comments:							

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APPENDIX A (reverse side)

IEPA Continuous Monitoring Log Sheet

DEPLOYMENT

Map of Station

Did you include:

North Arrow

Bridge and distance to sonde

Flow Direction

Location of Sonde

2 landmarks and distance to sonde from each

Cross-section Profile of Wetted Channel

SHOW LOCATION OF SONDE